

Multinational Firms and Trade in Latin America

Six Stylized Facts on Ownership and Exports

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Abstract

We match the customs export records of nine Latin American countries to the ownership records of Orbis and Dun & Bradstreet and classify every exporter as a foreign multinational affiliate, a domestic multinational affiliate or a local firm. Multinational affiliates account for 61 percent of the region's exports, most of it foreign owned. Foreign affiliates concentrate in complex and differentiated products, domestic ones in primary goods. The parents come from a few countries, and a small number of large groups accounts for most of the value. Product–destination pairs with more multinationals export more, and distance reduces the exports of multinationals less than those of other firms. US groups are the only large investors that ship a sizeable share of their affiliates' exports to their own country: 22 percent, against less than 3 percent for British, Dutch or Swiss groups.

Keywords: Multinational Firms, Foreign Direct Investment, Exports, Ownership, Firm-Level Data, Latin America

JEL codes: F14, F23, D22, L25

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1 Introduction

Customs records show what a country exports and where, but not who owns the exporter. In Latin America many exporters are affiliates of groups headquartered abroad, so this is a large gap. Foreign direct investment statistics do not fill it: they show where capital goes, not which firms export which products to which markets.

In this paper we match the customs export records of nine countries (Argentina, Chile, Colombia, Costa Rica, the Dominican Republic, El Salvador, Paraguay, Peru and Uruguay) to the ownership records of Orbis and Dun & Bradstreet. Every exporter is classified as a foreign multinational affiliate, a domestic multinational affiliate or a local firm, and the exports of foreign affiliates can be traced to the country of their parent.

The data show six facts. First, multinational affiliates account for 61 percent of the region's exports, from 47 percent in Peru to 74 percent in Colombia, and most of it is foreign owned. Second, foreign affiliates concentrate in complex and differentiated products and domestic multinationals in primary goods, especially within manufacturing. Third, the parents come from a few countries: the United States, the United Kingdom and Canada account for half of foreign-affiliate exports. Fourth, parents with more than 100 affiliates worldwide are 9 percent of all parents and account for 56 percent of foreign-affiliate exports. Fifth, product–destination pairs with more multinationals export more, and the relationship is about three times stronger for foreign than for domestic affiliates, also for the exports of local firms. Sixth, distance reduces the exports of multinationals less than those of other firms.

The United States deserves separate attention because it is both the largest parent country and the largest destination. Affiliates of US groups account for 10.8 percent of the exports of the nine countries, the United States buys 18.4 percent of them, and the two overlap in 2.5 percent. US groups are the only large investors that ship a sizeable share of their affiliates' exports to their own country: 22.3 percent, against 0.6 percent for British groups, 2.2 percent for Dutch groups and almost none for Swiss or Liechtenstein groups. These shipments are concentrated in Central America and the Caribbean, where US-parent affiliates account for 41 percent of El Salvador's exports to the United States, 40 percent of Costa Rica's and 26 percent of the Dominican Republic's. Groups from other countries follow different patterns. British and Liechtenstein groups, mostly in mining, sell to Asia, and German and Japanese groups export from Argentina to Brazil.

Contribution to the literature: to be written.

Section 2 reviews the literature, Section 3 describes the data and Section 4 presents the six facts. Section 5 looks at products, Section 6 outlines the model and Section 7 concludes.

2 Related Literature

To be written. Three strands: firm-level evidence on multinational production and trade; ownership, market power and trade policy; and evidence on multinational firms in Latin America.

3 Data

3.1 Customs data

The export records come from the customs authorities of the nine countries. Each record gives the value exported (FOB, US dollars) by a firm, of an HS6 product (2007 revision), to a destination, in a year. Firms are identified by their tax number and name. The records cover all merchandise exports in the years available for each country: Argentina 2011–2019, Chile 2009–2022, Colombia 2010–2021, Costa Rica 2010–2019, the Dominican Republic 2012–2019, El Salvador 2006–2018, Paraguay 2012–2020, Peru 2010–2019 and Uruguay 2010–2019.

3.2 Ownership data and matching

Ownership comes from Orbis (Bureau van Dijk) and Dun & Bradstreet, which report for each company its global ultimate owner and the owner’s country. We combine the two sources into one file with the group, the country of the parent and the number of affiliates the group has worldwide.

Exporters are matched to this file in four steps: by tax number where the corporate records report it; by company name within each country, using a character-based similarity measure, with each candidate pair reviewed by a large language model; by company name across all countries; and, for the 500 largest unmatched exporters of each country, by hand, using company websites and business registries. The tax number delivers about 60 percent of matched export value. After matching, the parent’s country is known for 92.5 percent of the export value of foreign affiliates.

3.3 Definitions

An exporter found in Orbis or Dun & Bradstreet is a multinational affiliate (MNE). It is a *domestic* MNE when its ultimate parent is located in the exporting country and a *foreign* MNE otherwise; matched exporters without a recorded parent country are counted as foreign. Exporters not found in either database are *local* firms. Because the match relies on ownership links, a group’s head company exporting from its own country cannot be identified and appears among local firms.

Products are classified with the Product Complexity Index of Hausmann and Hidalgo, the technology classes of Lall (2000), the Rauch (1999) classification, and the end-use classes of BEC Rev. 4. Sectors are defined on HS chapters: agriculture 01–24, mining and fuels 25–27 and 71, manufacturing 28–97 excluding 71. Distance comes from the CEPII gravity database.

3.4 Sample

Table 1 describes the sample by country and Table 2 summarizes it. The nine countries have 165,543 exporters, of which 31,350 are matched to a corporate record. They export 5,930 HS6 products to 253 destinations. The matched exporters belong to 17,094 groups with parents in 165 countries.

Table 1: Sample Coverage, by Exporting Country

Origin	Years	Exporters	Matched	HS6 lines	Destinations	Records	Value (US\$bn/yr)
ARG	2011–2019	22,980	6,687	4,657	202	1,105,127	66.4
COL	2010–2021	47,711	7,711	4,850	223	1,164,935	44.0
PER	2010–2019	27,286	4,696	4,600	215	845,003	41.4
CHL	2009–2022	17,430	5,776	4,673	205	589,227	34.2
CRI	2010–2019	14,816	1,387	4,452	197	530,375	10.5
DOM	2012–2019	15,832	1,671	3,970	206	329,133	8.6
PRY	2012–2020	3,220	999	2,801	168	60,188	7.3
URY	2010–2019	5,828	1,465	3,611	208	121,452	7.0
SLV	2006–2018	10,440	958	4,310	183	362,774	5.0
All nine	2006–2022	165,543	31,350	5,930	253	5,108,214	224.5

Notes: Exporters are distinct tax identifiers within each country; matched exporters are those found in Orbis or Dun & Bradstreet. Records are firm–destination–HS6–year observations. Values are annual averages in US\$ billion.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 2: Sample Summary

Statistic	Value
Exporting countries	9
Years	2006–2022
Records (firm $\times$ destination $\times$ HS6 $\times$ year)	5,108,214
Exporters	165,543
matched to Orbis / Dun & Bradstreet	31,350
HS6 product lines	5,930
Destination markets	253
Parent countries of foreign multinationals	165
Multinational groups	17,094
Export value, all years (US\$ billion)	2,394
foreign multinationals	46.3%
domestic multinationals	14.1%
local firms	39.5%
Foreign-MNE value with a recorded parent country	92.5%

Notes: Nine countries, all available years pooled. Shares are percent of total export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

4 Six Stylized Facts

4.1 Fact 1: Multinationals Account for a Large Share of Exports

Figure 1 shows the share of each country's exports accounted for by multinational affiliates. It ranges from 47 percent in Peru to 74 percent in Colombia and is 61 percent for the nine countries together. Foreign affiliates account for 46 percent of total exports and domestic multinationals for 14 percent. Foreign affiliates export more than domestic multinationals in every country except Colombia, where the state oil company and its subsidiaries are domestic multinationals. Chile, with copper and pulp, is the other country with a large domestic share.

Figures 2 and 3 and Table 3 separate the two roles of the United States. The United States buys 18.4 percent of the region's exports and US-parent affiliates ship 10.8 percent, but only 2.5 percent is both. US-parent affiliates account for 13.6 percent of the region's exports to the United States, and for much more in El Salvador, Costa Rica and the Dominican Republic.

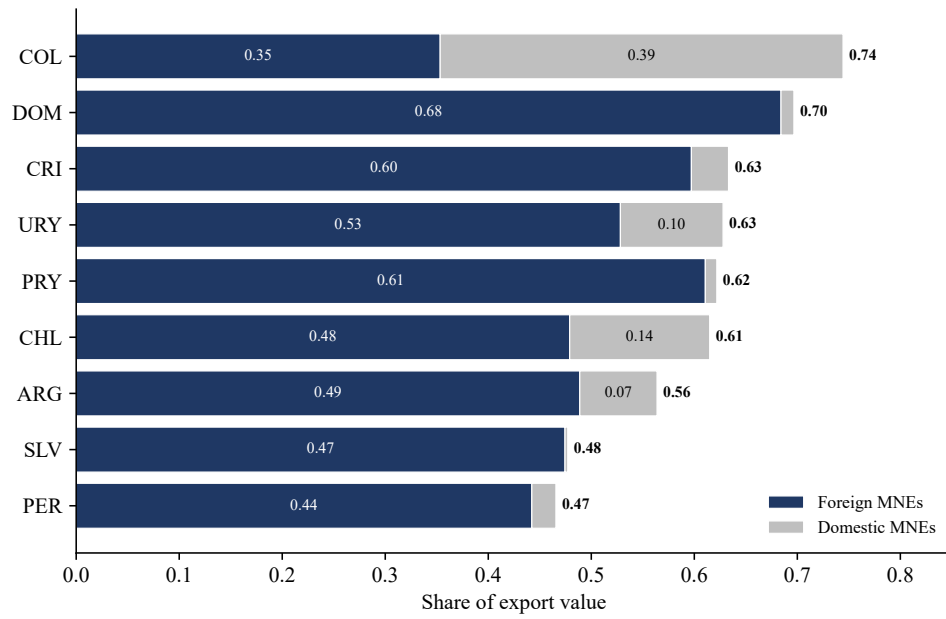


Figure 1: Multinational Share of Export Value, by Exporting Country

Notes: Shares of export value, all available years pooled. The figure at the end of each bar is the total MNE share.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

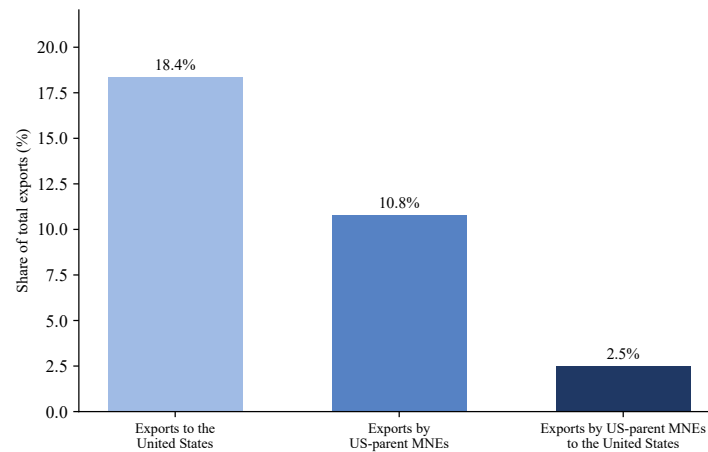


Figure 2: Exports to the United States and Exports by US-Parent Multinationals

Notes: Shares of the total exports of the nine countries. The third bar is exports by US-parent MNEs shipped to the United States.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

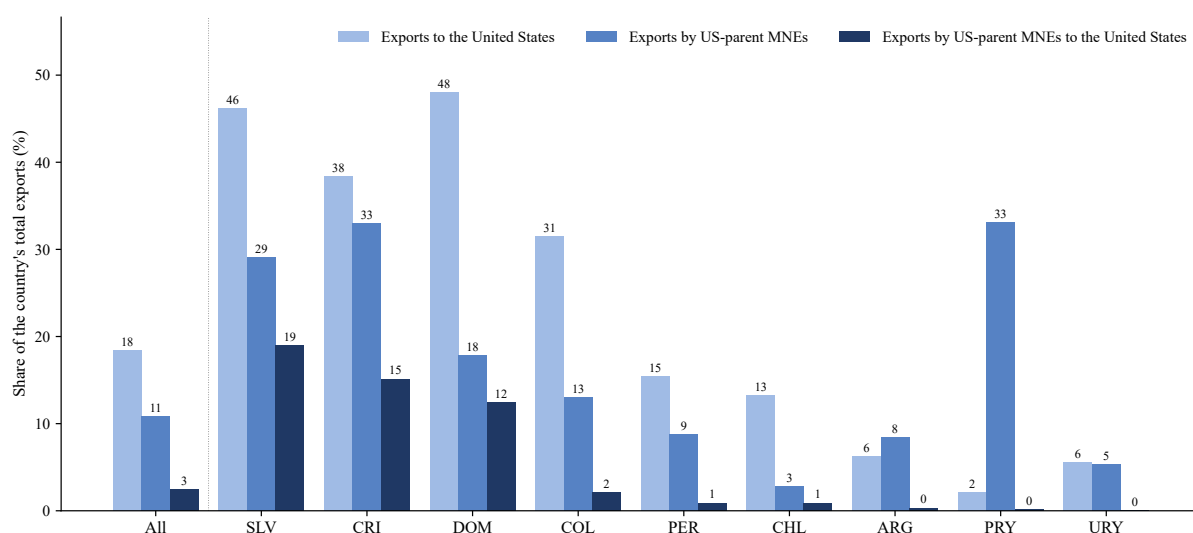


Figure 3: Exports to the United States and Exports by US-Parent Multinationals, by Exporting Country

Notes: Shares of each country's total exports.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 3: Exports to the United States and Exports by US-Parent Multinationals, by Exporting Country

Origin	(1) % of exports going to the USA	(2) % of exports carried by US-parent MNEs	(3) % of exports carried by US-parent MNEs and going to the USA	(4) = (3)/(1): US-parent share of the exports going to the USA (%)	Total exports (US\$bn/yr)
All	18.4	10.8	2.5	13.6	224.5
SLV	46.1	29.1	19.0	41.3	5.0
CRI	38.3	33.0	15.1	39.5	10.5
DOM	48.0	17.8	12.5	26.0	8.6
COL	31.5	13.1	2.1	6.8	44.0
PER	15.4	8.8	0.9	6.0	41.4
CHL	13.3	2.8	0.9	6.9	34.2
ARG	6.3	8.4	0.3	5.5	66.4
PRY	2.1	33.1	0.2	8.1	7.3
URY	5.5	5.3	0.0	0.4	7.0

Notes: Columns (1)–(3): percent of each country's total exports. Column (4): percent of exports to the United States shipped by US-parent MNEs. Countries sorted by column (3). Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

4.2 Fact 2: Foreign Multinationals Specialize in Complex Products, Domestic Multinationals in Primary Goods

Figure 4 sorts products by sophistication. The foreign share is 0.46 in the least complex quintile of the Product Complexity Index and 0.65 in the most complex one, and it is highest in high- and medium-technology manufactures. Domestic multinationals are concentrated in primary and resource-based products. The Rauch classification gives the same ordering (Table 4 and Figure 5): foreign affiliates account for 56 percent of exports of differentiated products and 34 percent of homogeneous products, while the domestic shares are 6 and 23 percent. By end use, the foreign share is 65 percent in capital goods, 48 percent in intermediate goods and 44 percent in consumption goods (Table 5). Table 6 shows the same patterns in regressions at the origin–destination–product–year level. The pattern is clearest within manufacturing (Figure 6); it is weak within agriculture and absent within mining.

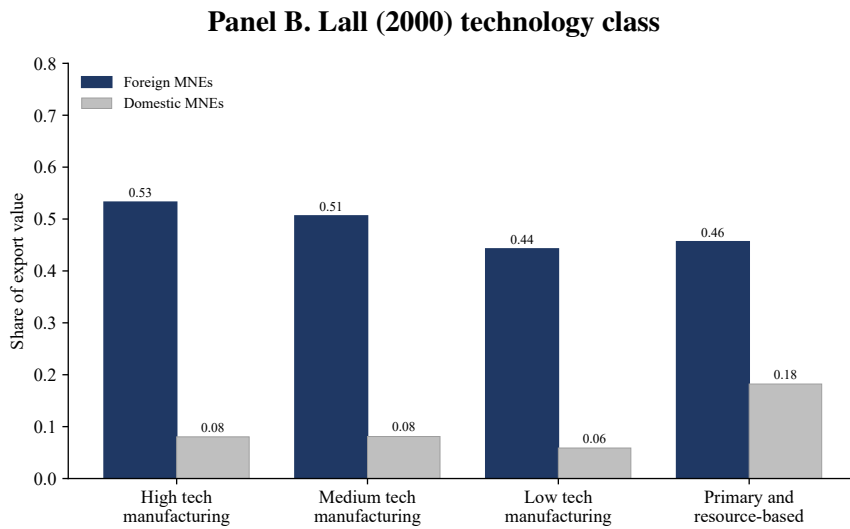
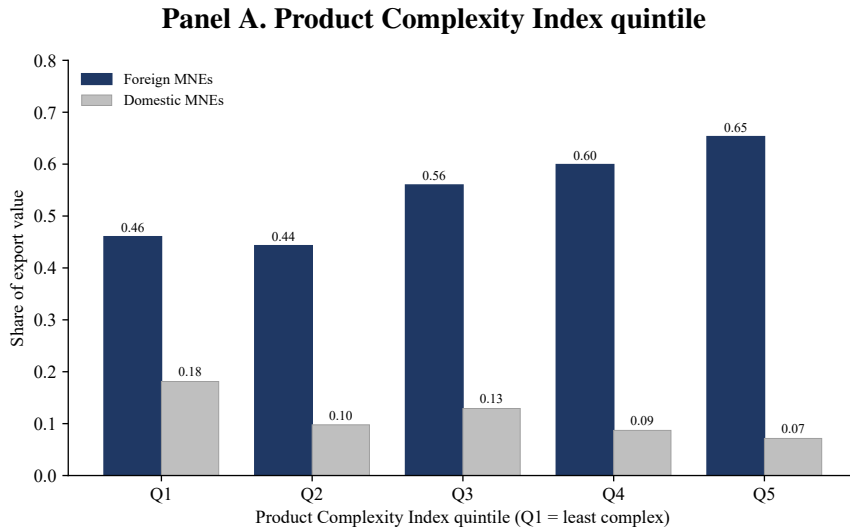


Figure 4: Multinational Share of Export Value, by Product Sophistication

Notes: Shares of export value, all available years pooled. Panel A: Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex). Panel B: Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 4: Multinational Share of Export Value, by Rauch Product Class

Rauch (1999) class	Foreign	Domestic	All MNEs	HS6 products	Value (US\$bn/yr)
Differentiated	0.563	0.058	0.621	2,712	44.3
Reference-priced	0.549	0.082	0.631	1,636	63.7
Homogeneous (exchange)	0.343	0.228	0.571	319	97.6

Notes: Shares of export value, all available years pooled. Rauch (1999) conservative classification: homogeneous goods are traded on organized exchanges, reference-priced goods have a price quoted in trade publications, and differentiated goods have neither. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

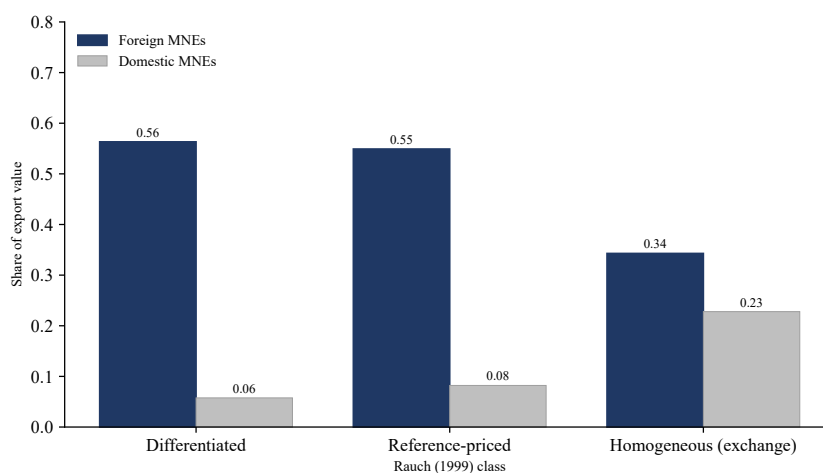


Figure 5: Multinational Share of Export Value, by Rauch Product Class

Notes: Shares of export value, all available years pooled. Rauch (1999) conservative classification: homogeneous goods are traded on organized exchanges, reference-priced goods have a price quoted in trade publications, and differentiated goods have neither.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 5: Multinational Share of Export Value, by End Use

BEC end use	Foreign	Domestic	All MNEs	HS6 products	Value (US\$bn/yr)
Intermediate	0.476	0.155	0.631	3,090	155.0
Consumption	0.438	0.082	0.520	1,201	46.1
Capital	0.647	0.056	0.703	659	9.3

Notes: Shares of export value, all available years pooled. End use from BEC Rev. 4; HS6 products without a BEC correspondence are excluded. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 6: Product Sophistication and Multinational Shares

	PCI		Lall		Rauch	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Dependent variable: share of all MNEs</i>						
Complexity (PCI)	-0.0004 (0.0050)	0.0052 (0.0045)				
Lall: high tech (vs primary and resource-based)			-0.0288** (0.0142)	-0.0347** (0.0142)		
Lall: medium tech			-0.0256*** (0.0081)	-0.0111 (0.0074)		
Lall: low tech			-0.0801*** (0.0165)	-0.0797*** (0.0171)		
Rauch: differentiated (vs homogeneous)					0.0819*** (0.0112)	0.0880*** (0.0104)
Rauch: reference-priced					0.1038*** (0.0103)	0.1132*** (0.0104)
Observations	1,671,802	1,670,186	1,788,929	1,787,370	1,667,472	1,665,924
<i>Panel B. Dependent variable: share of foreign MNEs</i>						
Complexity (PCI)	0.0385*** (0.0052)	0.0280*** (0.0050)				
Lall: high tech (vs primary and resource-based)			0.0330** (0.0135)	0.0140 (0.0129)		
Lall: medium tech			0.0400*** (0.0080)	0.0517*** (0.0071)		
Lall: low tech			-0.0000 (0.0118)	-0.0050 (0.0087)		
Rauch: differentiated (vs homogeneous)					0.2088*** (0.0082)	0.1914*** (0.0079)
Rauch: reference-priced					0.1783*** (0.0099)	0.1659*** (0.0099)
Observations	1,671,802	1,670,186	1,788,929	1,787,370	1,667,472	1,665,924
<i>Panel C. Dependent variable: share of domestic MNEs</i>						
Complexity (PCI)	-0.0389*** (0.0058)	-0.0227*** (0.0045)				
Lall: high tech (vs primary and resource-based)			-0.0618*** (0.0143)	-0.0487*** (0.0137)		
Lall: medium tech			-0.0657*** (0.0087)	-0.0628*** (0.0075)		
Lall: low tech			-0.0801*** (0.0148)	-0.0747*** (0.0155)		
Rauch: differentiated (vs homogeneous)					-0.1270*** (0.0101)	-0.1034*** (0.0084)
Rauch: reference-priced					-0.0745*** (0.0086)	-0.0527*** (0.0075)
Observations	1,671,802	1,670,186	1,788,929	1,787,370	1,667,472	1,665,924
Origin × year and destination × year FE	✓		✓		✓	
Origin × dest. × year FE		✓		✓		✓

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: share of the cell’s export value shipped by each type of MNE. Omitted categories: primary and resource-based products (Lall) and homogeneous products (Rauch). Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

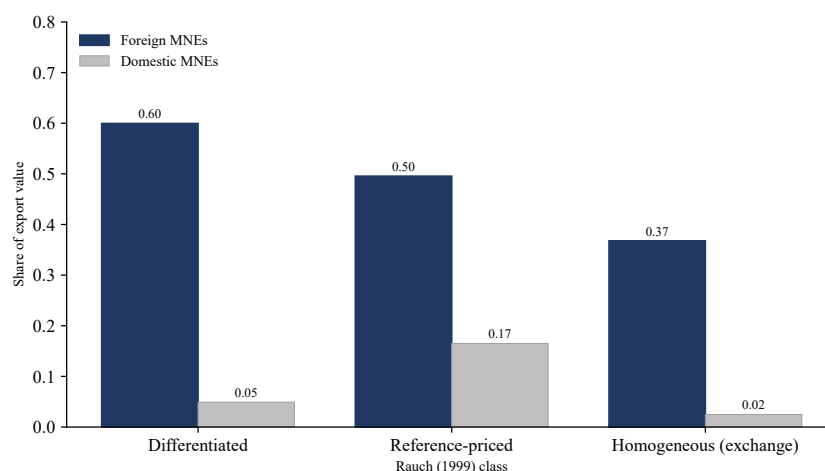


Figure 6: Manufacturing: Multinational Share of Export Value, by Rauch Product Class

Notes: Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Rauch (1999) conservative classification: homogeneous goods are traded on organized exchanges, reference-priced goods have a price quoted in trade publications, and differentiated goods have neither.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

4.3 Fact 3: Multinationals from a Small Set of Countries Dominate Exports

Figure 7 ranks parent countries by their share of foreign-affiliate exports. The United States (23 percent), the United Kingdom (19 percent) and Canada (8 percent) account for half, and the ten largest parent countries for more than three quarters. Table 7 shows where the affiliates of each of the ten largest parent countries sell; its diagonal is the share shipped to the parent's own country. The share is 22.3 percent for US groups, 19.5 percent for Canadian groups and 18.6 percent for Brazilian groups, but less than 3 percent for British, Dutch or Swiss groups. Figure 8 shows the same pattern from the side of the main destination markets.

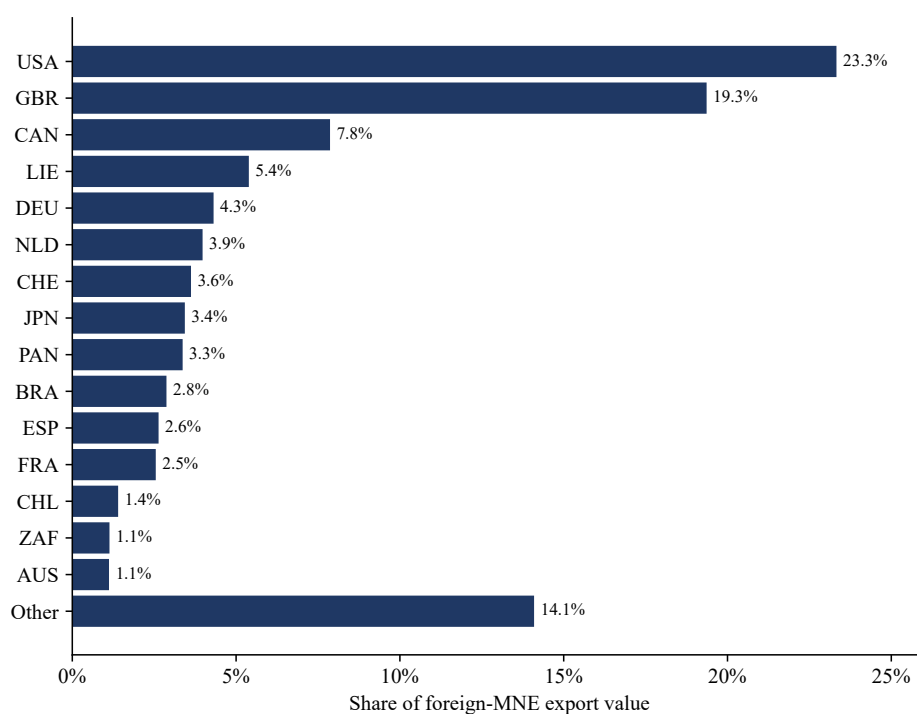


Figure 7: Foreign-Multinational Exports, by Parent Country

Notes: Percent of foreign-MNE export value, by country of the ultimate parent.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 7: Foreign-Multinational Exports to the Parent Countries, by Parent Country

Parent / Destination	USA	GBR	CAN	LIE	DEU	NLD	CHE	JPN	PAN	BRA	Other destinations	Total
USA	22.3	1.0	0.5	0.0	0.6	3.5	1.7	3.8	2.7	7.8	56.2	100.0
GBR	5.6	0.6	3.3	0.0	3.6	3.2	0.6	7.1	0.2	3.9	71.8	100.0
CAN	6.6	2.6	19.5	0.0	3.5	0.9	12.9	5.3	2.6	4.2	41.8	100.0
LIE	3.7	0.4	0.0	0.0	0.9	1.8	0.2	18.7	0.1	3.7	70.5	100.0
DEU	14.8	0.3	1.1	0.0	8.3	1.0	0.0	1.1	0.8	31.9	40.6	100.0
NLD	3.8	0.5	0.2	0.0	0.4	2.2	0.0	0.4	0.4	26.9	65.2	100.0
CHE	5.0	1.0	0.4	0.0	0.7	2.0	0.3	0.5	1.7	6.8	81.6	100.0
JPN	8.7	0.3	0.5	0.0	0.2	0.9	0.0	8.9	1.0	45.0	34.5	100.0
PAN	9.5	0.8	0.6	0.0	1.7	2.0	0.2	1.8	1.4	1.2	80.7	100.0
BRA	14.1	0.5	0.6	0.0	2.3	2.1	2.2	1.6	0.6	18.6	57.4	100.0
Other parents	19.5	0.9	1.6	0.0	2.2	2.1	1.2	2.2	1.7	11.7	56.8	100.0

Notes: Percent of each parent country's exports, by destination (rows add to 100). Destinations are the ten largest parent countries; the diagonal is the share shipped to the parent's own country.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

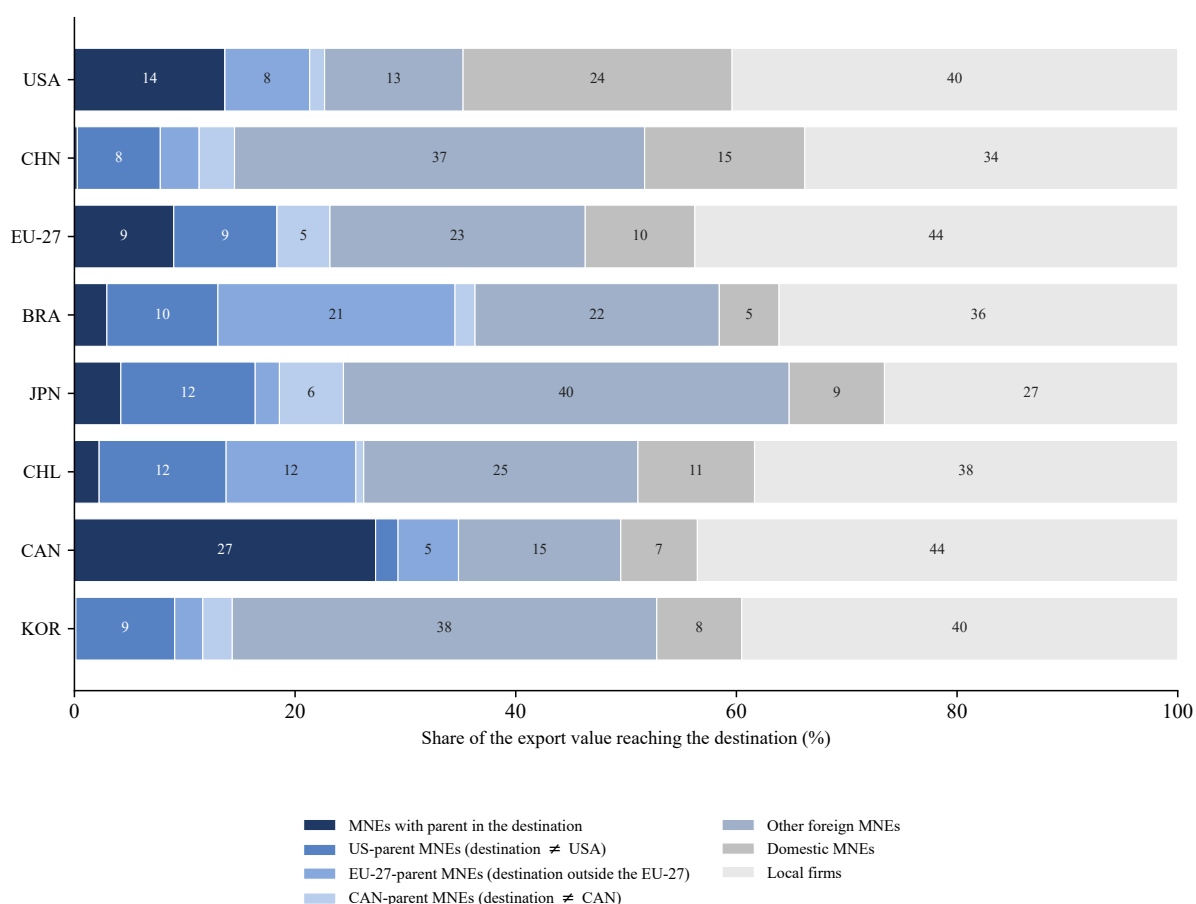


Figure 8: Ownership of Exports to the Main Destination Markets

Notes: Percent of the export value reaching each destination, by type of exporter. The first segment is exports by MNEs whose parent is located in that destination. EU-27 member states are pooled.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

4.4 Fact 4: A Small Set of Large Multinational Groups Accounts for Most Exports

Figure 9 groups parents by the number of affiliates their group has worldwide. Almost half of the parents have a single affiliate, but the 9 percent with more than 100 affiliates account for 56 percent of foreign-affiliate exports.

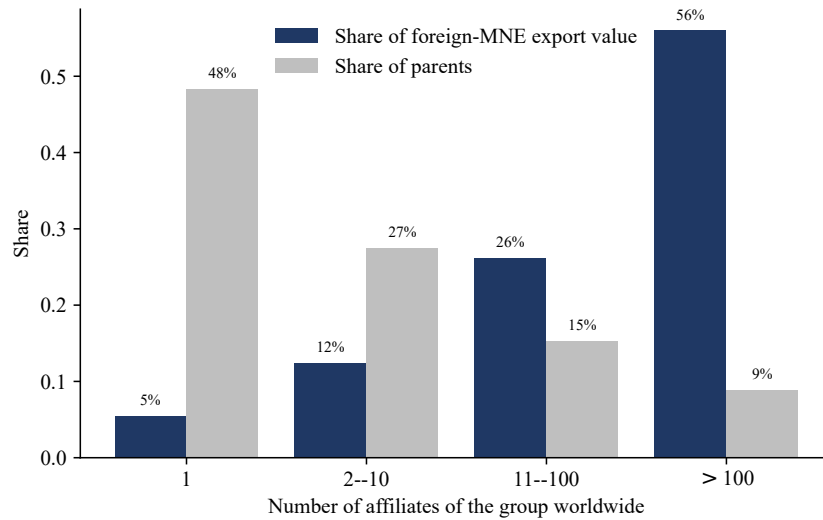


Figure 9: Foreign-Multinational Exports and Number of Parents, by Size of the Group's Global Network

Notes: Shares of foreign-MNE export value and of parents, by the number of affiliates the group has worldwide.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

4.5 Fact 5: Multinational Presence Is Associated with Higher Exports

Table 8 relates the exports of an origin–destination–product–year cell to the number of multinationals exporting in it (intensive margin) and to whether any does (extensive margin). Both relationships are positive and large, also for the exports of the local firms in the same cell (Panel B). Table 9 splits the intensive margin by type of multinational. The coefficient is 1.21 for foreign affiliates and 0.44 for domestic ones in Panel A, and 1.13 and 0.47 in Panel B. On the extensive margin the two coefficients are almost equal (Table A.9).

Table 8: Multinational Presence and Exports

	Intensive: ln(# MNE firms)			Extensive: any MNE present		
	(1)	(2)	(3)	(1)	(2)	(3)
<i>Panel A. Dependent variable: ln exports, all firms</i>						
ln(# MNE firms)	2.0823*** (0.0363)	2.0685*** (0.0368)	0.9414*** (0.0206)			
Any MNE present				1.4622*** (0.0394)	1.4591*** (0.0388)	0.6818*** (0.0165)
<i>Panel B. Dependent variable: ln exports, non-MNE firms</i>						
ln(# MNE firms)	1.3536*** (0.0318)	1.3406*** (0.0319)	0.2389*** (0.0178)			
Any MNE present				1.4168*** (0.0371)	1.3996*** (0.0367)	0.1311*** (0.0102)
Origin FE	✓			✓		
Destination FE	✓			✓		
Product FE	✓	✓		✓	✓	
Year FE	✓			✓		
Origin × year FE		✓			✓	
Destination × year FE		✓			✓	
Origin × dest. × product FE			✓			✓
Origin × dest. × year FE			✓			✓
Observations (Panel A)	1,008,472	1,008,185	883,545	1,829,852	1,829,626	1,633,319
Observations (Panel B)	475,856	475,514	429,882	1,297,322	1,297,111	1,137,550

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Intensive margin: ln number of MNEs, cells with at least one MNE. Extensive margin: indicator for at least one MNE. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 9: Multinational Presence and Exports: Intensive Margin, by Type of Multinational

	All MNEs		Foreign / domestic		Foreign by destination, domestic	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Dependent variable: ln exports, all firms</i>						
ln(# MNE firms)	2.0685*** (0.0368)	0.9414*** (0.0206)				
ln(# foreign MNEs)			1.2129*** (0.0285)	0.4506*** (0.0124)		
ln(# foreign MNEs, parent-country destination)					0.3970*** (0.0856)	0.1448*** (0.0370)
ln(# foreign MNEs, other destinations)					0.9465*** (0.0536)	0.3133*** (0.0382)
ln(# domestic MNEs)			0.4382*** (0.0259)	0.2214*** (0.0151)	0.3770*** (0.0514)	0.1555*** (0.0315)
<i>Panel B. Dependent variable: ln exports, non-MNE firms</i>						
ln(# MNE firms)	1.3406*** (0.0319)	0.2389*** (0.0178)				
ln(# foreign MNEs)			1.1331*** (0.0375)	0.2413*** (0.0217)		
ln(# foreign MNEs, parent-country destination)					0.2934*** (0.0882)	0.1945*** (0.0527)
ln(# foreign MNEs, other destinations)					1.0330*** (0.0607)	0.1951*** (0.0648)
ln(# domestic MNEs)			0.4707*** (0.0341)	0.1250*** (0.0218)	0.4142*** (0.0633)	0.0946* (0.0552)
Product FE	✓		✓		✓	
Origin × year FE	✓		✓		✓	
Destination × year FE	✓		✓		✓	
Origin × dest. × product FE		✓		✓		✓
Origin × dest. × year FE		✓		✓		✓
Observations (Panel A)	1,008,185	883,545	113,422	98,223	9,197	7,524
Observations (Panel B)	475,514	429,882	93,010	81,563	8,780	7,255

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: ln number of MNEs of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

4.6 Fact 6: Distance Is a Weaker Barrier for Multinationals

Table 10 estimates how firm exports fall with distance. The elasticity is -0.16 for local firms and about 0.05 smaller in absolute value for multinationals. Table 11 shows that the difference is larger for foreign affiliates (0.049) than for domestic ones (0.032), and largest for foreign affiliates exporting to their parent’s country (0.064).

Table 10: Distance and Firm Exports: Multinationals by Presence at the Destination

Dep. var.: ln exports	(1)	(2)	(3)	(4)
ln distance	-0.1639*** (0.0454)	-0.1848*** (0.0460)	-0.1632*** (0.0452)	-0.1838*** (0.0459)
ln distance × MNE	0.0461*** (0.0038)	0.0458*** (0.0038)		
ln distance × MNE, present at destination			0.0689*** (0.0070)	0.0682*** (0.0070)
ln distance × MNE, not present at destination			0.0393*** (0.0032)	0.0391*** (0.0032)
Origin FE	✓		✓	
Destination FE	✓		✓	
Year FE	✓		✓	
Origin × year FE		✓		✓
Destination × year FE		✓		✓
Product FE	✓	✓	✓	✓
Observations	5,081,547	5,081,360	5,081,547	5,081,360

Notes: Observations are firm–destination–HS6–year cells. Dependent variable: ln exports. Non-MNE firms are the omitted group. Present: the group has an affiliate in the destination or the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 11: Distance and Firm Exports: Foreign and Domestic Multinationals

Dep. var.: ln exports	(1)	(2)	(3)	(4)
ln distance	-0.1645*** (0.0454)	-0.1853*** (0.0460)	-0.1644*** (0.0454)	-0.1851*** (0.0460)
ln distance × foreign MNE	0.0487*** (0.0041)	0.0483*** (0.0040)		
ln distance × foreign MNE, parent-country destination			0.0642*** (0.0159)	0.0638*** (0.0161)
ln distance × foreign MNE, other destination			0.0477*** (0.0035)	0.0473*** (0.0035)
ln distance × domestic MNE	0.0318*** (0.0042)	0.0318*** (0.0042)	0.0317*** (0.0042)	0.0317*** (0.0042)
Origin FE	✓		✓	
Destination FE	✓		✓	
Year FE	✓		✓	
Origin × year FE		✓		✓
Destination × year FE		✓		✓
Product FE	✓	✓	✓	✓
Observations	5,081,547	5,081,360	5,081,547	5,081,360

Notes: Observations are firm–destination–HS6–year cells. Dependent variable: ln exports. Non-MNE firms are the omitted group. Columns (3)–(4) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

5 Products

Figure 10 shows the twenty largest export products and who exports them. Crude petroleum is mostly exported by domestic multinationals and copper concentrates by foreign ones. Figure 11 splits the foreign share by parent country: British groups in copper and coal, Canadian groups in gold, US groups in soybeans, maize and apparel, German groups in trucks. Table 12 gives the same information by HS section, and Tables A.15 to A.18 in the appendix break down agriculture, manufacturing and mining.

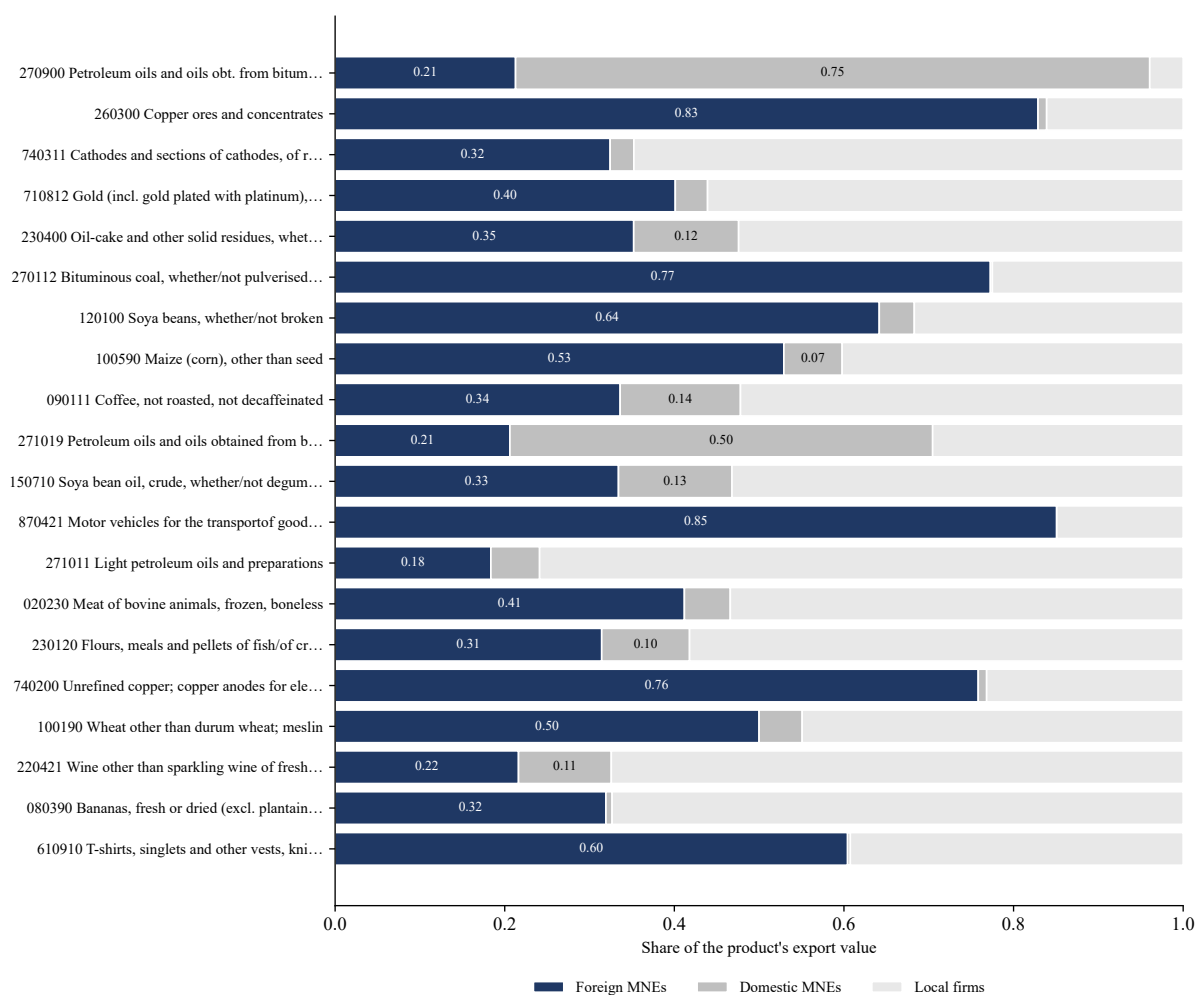


Figure 10: Ownership of the Twenty Largest Export Products

Notes: Shares of each product's export value. Twenty largest HS6 products by export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

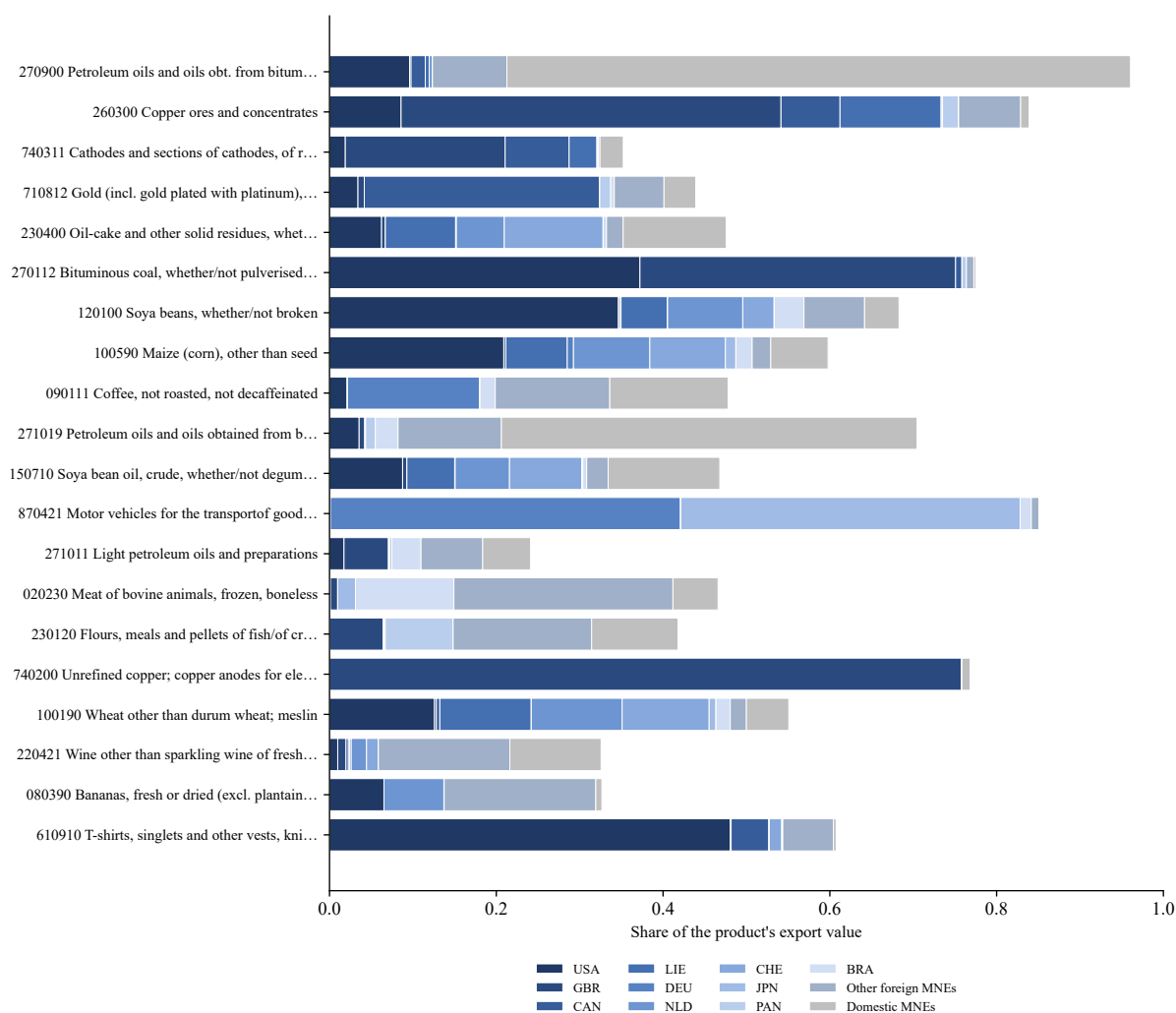


Figure 11: Ownership of the Twenty Largest Export Products, by Parent Country

Notes: Shares of each product's export value. Foreign-MNE share split by country of the ultimate parent; the unfilled part of each bar is local firms.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 12: Exports and Multinational Shares, by HS Section

HS section	\$bn/yr	For.	Domestic	Local	Lead parent
V Mineral products	57.1	51	29	19	GBR (0.42)
II Vegetable products	33.5	42	7	51	USA (0.33)
IV Prepared foodstuffs, beverages, tobacco	25.9	37	10	53	USA (0.21)
XV Base metals	19.6	41	3	56	GBR (0.44)
XIV Precious metals and stones	15.3	41	4	56	CAN (0.65)
I Live animals; animal products	12.5	34	14	51	SAU (0.19)
VI Chemicals	11.6	49	14	37	USA (0.25)
XVII Transport equipment	9.7	78	4	18	DEU (0.30)
XI Textiles and apparel	6.9	49	3	48	USA (0.62)
III Fats and oils	6.9	31	10	59	CHE (0.26)
XVI Machinery and electrical equipment	6.2	63	4	33	USA (0.41)
VII Plastics and rubber	5.8	51	8	42	USA (0.34)
X Pulp, paper	3.4	54	28	18	PAN (0.54)
XVIII Optical, precision instruments	3.4	72	8	19	USA (0.66)
IX Wood and articles	1.9	27	34	39	BHS (0.24)
VIII Hides, skins, leather	1.6	40	12	49	BGR (0.19)
XIII Stone, ceramics, glass	0.9	49	4	46	USA (0.32)
XX Miscellaneous manufactures	0.9	53	7	40	USA (0.48)
Unclassified	0.5	49	29	23	USA (0.39)
XII Footwear, headgear	0.5	48	1	50	USA (0.66)
XXI Works of art	0.1	58	1	41	BGR (0.94)
XIX Arms	0.0	1	0	99	CHE (0.82)
Special / unclassified (98-99)	0.0	13	1	87	USA (0.57)
All sections	224.5	46	14	40	

Notes: Foreign, domestic and local: percent of each section's export value. Leading parent: largest parent country and its share of the section's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

6 Theoretical Framework

To be written. The model is developed in the companion theory paper.

7 Concluding Remarks

To be written.

References

To be completed.

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A Additional Tables and Figures

Fact 1

Table A.1: Multinational Share of Export Value, by Exporting Country

Origin	Foreign	Domestic	All MNEs	Value (US\$bn/yr)
COL	0.353	0.391	0.744	44.0
DOM	0.684	0.013	0.697	8.6
CRI	0.597	0.036	0.633	10.5
URY	0.528	0.100	0.628	7.0
PRY	0.611	0.011	0.622	7.3
CHL	0.479	0.136	0.615	34.2
ARG	0.489	0.075	0.564	66.4
SLV	0.474	0.003	0.477	5.0
PER	0.442	0.023	0.466	41.4
All	0.463	0.141	0.605	224.5

Notes: Shares of each country's export value, all available years pooled. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

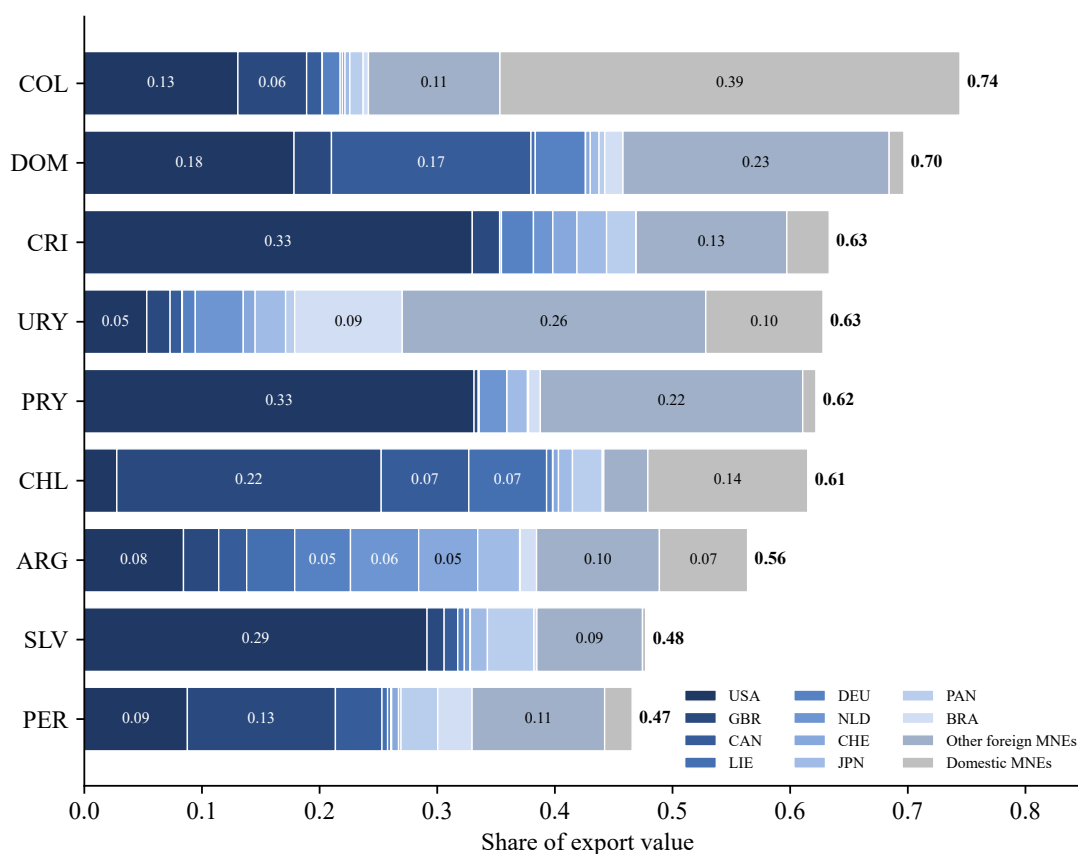


Figure A.1: Multinational Share of Export Value, by Exporting Country and Parent Country

Notes: Shares of each country's export value. Foreign MNEs split by country of the ultimate parent; the ten largest parent countries are shown separately.

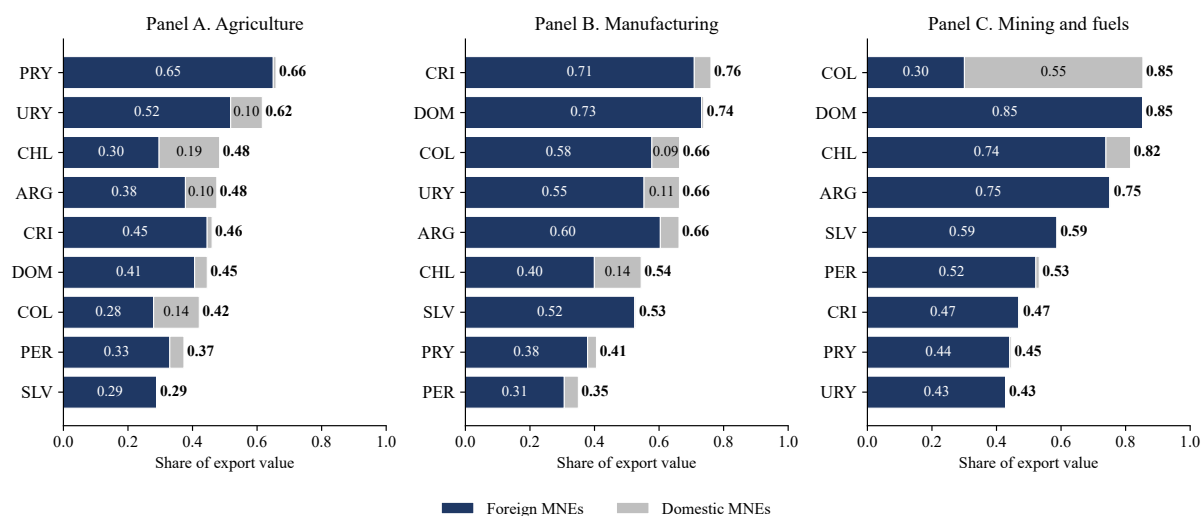
Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.2: Multinational Share of Export Value, by Exporting Country and Parent Country

Origin	USA	GBR	CAN	LIE	DEU	NLD	CHE	JPN	PAN	BRA	Other	Domestic	All MNEs	Value (US\$bn/yr)
COL	0.131	0.058	0.013	0.000	0.015	0.002	0.002	0.004	0.011	0.005	0.112	0.391	0.744	44.0
DOM	0.178	0.032	0.170	0.004	0.043	0.000	0.004	0.007	0.005	0.015	0.226	0.013	0.697	8.6
CRI	0.330	0.023	0.002	0.000	0.027	0.016	0.020	0.025	0.025	0.000	0.128	0.036	0.633	10.5
URY	0.053	0.020	0.010	0.000	0.011	0.041	0.010	0.026	0.008	0.091	0.258	0.100	0.628	7.0
PRY	0.331	0.004	0.000	0.000	0.000	0.023	0.000	0.017	0.001	0.010	0.223	0.011	0.622	7.3
CHL	0.028	0.225	0.074	0.066	0.005	0.000	0.005	0.012	0.025	0.001	0.037	0.136	0.615	34.2
ARG	0.084	0.030	0.024	0.041	0.047	0.058	0.050	0.036	0.001	0.014	0.104	0.075	0.564	66.4
SLV	0.291	0.015	0.012	0.000	0.006	0.005	0.000	0.014	0.040	0.003	0.090	0.003	0.477	5.0
PER	0.088	0.126	0.040	0.005	0.003	0.001	0.006	0.002	0.031	0.029	0.113	0.023	0.466	41.4

Notes: Shares of each country's export value. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure A.2: Multinational Share of Export Value, by Exporting Country and Sector**

Notes: Shares of each country's export value in the sector. Panel A: agriculture (HS 01–24). Panel B: manufacturing (HS 28–97 excluding 71). Panel C: mining and fuels (HS 25–27 and 71).

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.3: Exports and Multinational Shares, by Sector

Sector	Total exports (US\$bn/yr)	MNE exports (US\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
Agriculture	78.9	37.8	38.1	9.5	USA 0.24, CHE 0.11, NLD 0.10
Mining & fuels	72.4	52.4	49.2	24.5	GBR 0.36, USA 0.21, CAN 0.16
Manufacturing	72.7	43.9	51.6	8.0	USA 0.25, GBR 0.14, DEU 0.09
Rest	0.5	0.4	47.9	28.0	USA 0.39, GBR 0.12, ESP 0.11
All	224.5	134.5	46.3	14.1	

Notes: Foreign and domestic: percent of the sector's export value. Rest: HS 98–99 and unclassified codes. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Fact 2

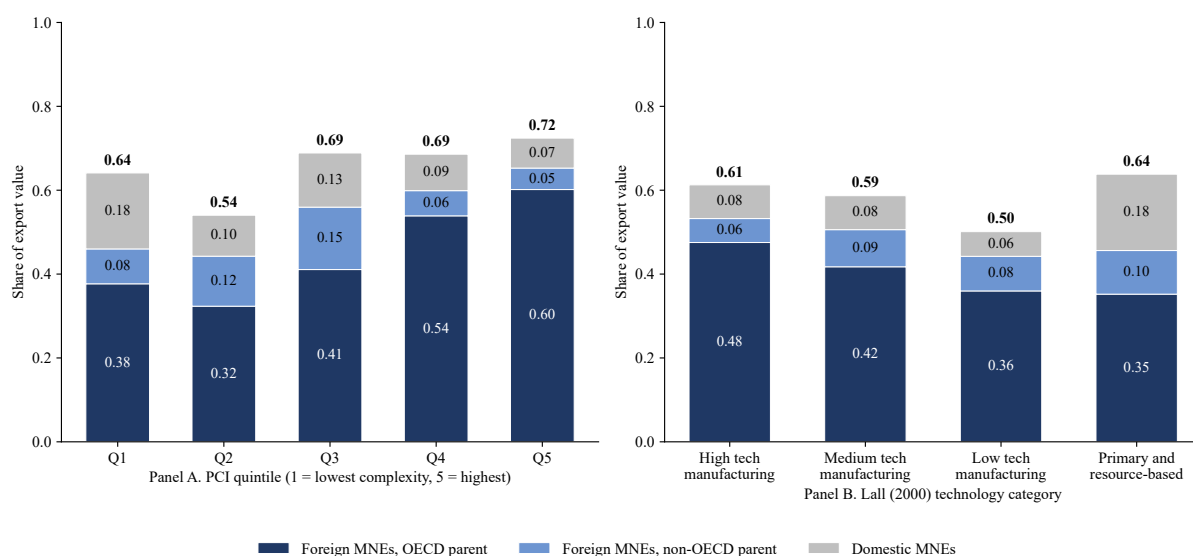


Figure A.3: Multinational Share of Export Value, by Product Sophistication and Location of the Parent

Notes: Shares of export value, all available years pooled. Foreign MNEs are split by whether the ultimate parent is in an OECD member country. Panel A: Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex). Panel B: Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

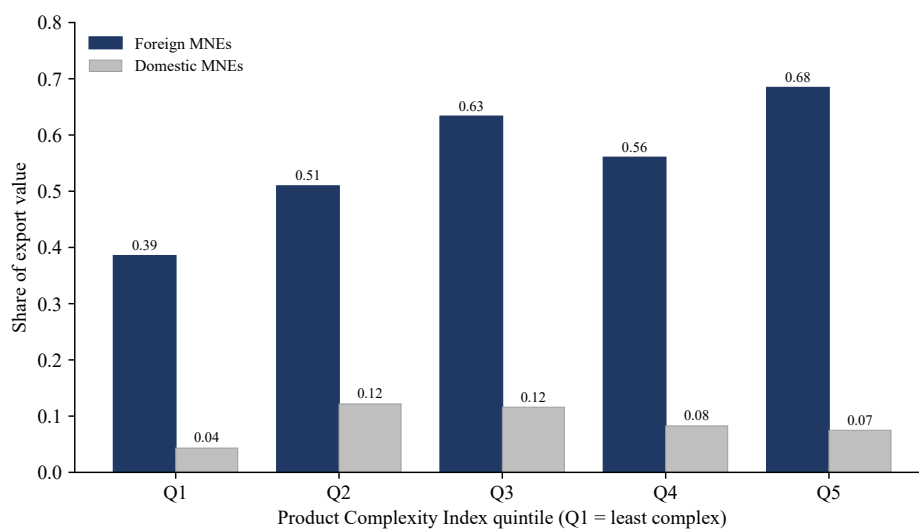


Figure A.4: Manufacturing: Multinational Share of Export Value, by Product Complexity Quintile

Notes: Shares of manufacturing export value, all available years pooled. Quintiles computed over manufacturing HS6 products.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Fact 3

Table A.4: Destination of Foreign-Multinational Exports, by Parent Country

Parent / Destination	CHN	USA	BRA	JPN	ESP	CHL	KOR	CAN	NLD	ARG	Other destinations	Total
USA	10.1	22.3	7.8	3.8	4.0	2.9	2.1	0.5	3.5	4.9	38.2	100.0
GBR	36.5	5.6	3.9	7.1	3.0	3.4	6.5	3.3	3.2	0.2	27.3	100.0
CAN	12.8	6.6	4.2	5.3	3.1	0.5	1.9	19.5	0.9	0.2	45.0	100.0
LIE	24.9	3.7	3.7	18.7	3.5	2.5	5.4	0.0	1.8	0.0	35.6	100.0
DEU	1.0	14.8	31.9	1.1	1.7	3.1	0.6	1.1	1.0	0.9	42.8	100.0
NLD	9.2	3.8	26.9	0.4	1.0	3.6	1.1	0.2	2.2	3.1	48.4	100.0
CHE	6.2	5.0	6.8	0.5	2.9	3.5	0.8	0.4	2.0	0.5	71.4	100.0
JPN	2.4	8.7	45.0	8.9	0.4	1.4	0.4	0.5	0.9	1.9	29.5	100.0
PAN	35.0	9.5	1.2	1.8	1.7	1.1	6.2	0.6	2.0	0.2	40.7	100.0
BRA	12.9	14.1	18.6	1.6	0.6	3.1	0.6	0.6	2.1	2.3	43.4	100.0
Other parents	9.5	19.5	11.7	2.2	3.7	4.0	1.6	1.6	2.1	3.7	40.3	100.0

Notes: Percent of each parent country's exports, by destination (rows add to 100). Ten largest parent countries and ten largest destinations by foreign-MNE export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.5: Destination of Multinational Exports, by Parent Region

Parent region / Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
Latin America	40.4	15.5	14.7	27.2	1.2	0.7	0.3	100.0
North America	31.3	23.6	20.0	23.1	1.7	0.1	0.1	100.0
Europe	28.0	11.7	17.8	38.4	2.9	1.0	0.1	100.0
Asia	51.0	18.7	11.7	17.0	1.1	0.2	0.3	100.0
Africa	10.9	6.6	26.9	53.2	2.2	0.0	0.2	100.0
Oceania	20.1	5.7	8.4	65.5	0.1	0.2	0.0	100.0
Rest of World	32.6	25.0	20.9	14.3	0.0	7.1	0.0	100.0
Domestic MNEs	25.9	32.9	11.1	27.0	2.3	0.5	0.3	100.0
Local firms	27.1	21.6	21.7	25.7	2.9	0.6	0.4	100.0

Notes: Percent of each group's exports, by destination region (rows add to 100). Domestic MNEs and local firms are shown for comparison.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

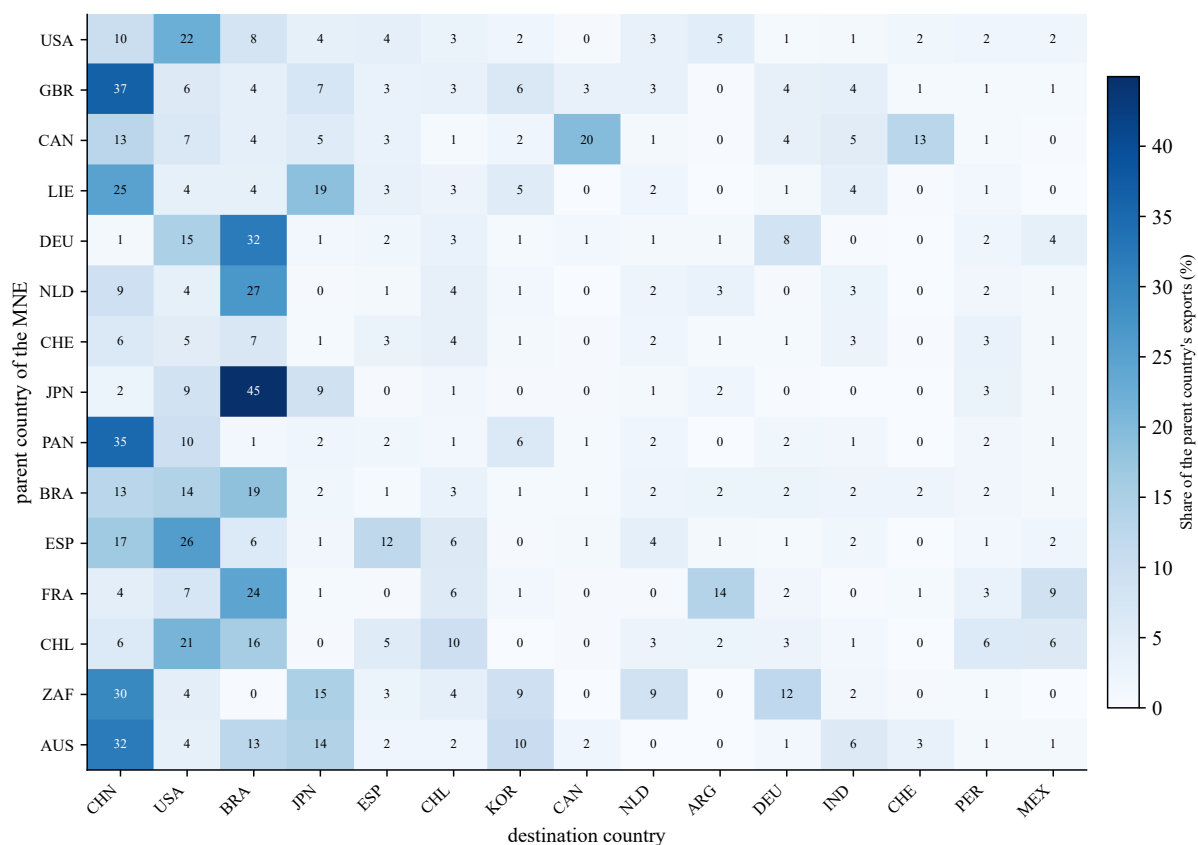


Figure A.5: Destination of Foreign-Multinational Exports: Fifteen Largest Parent Countries and Destinations

Notes: Percent of each parent country's exports going to each destination (rows add to 100 including other destinations).

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

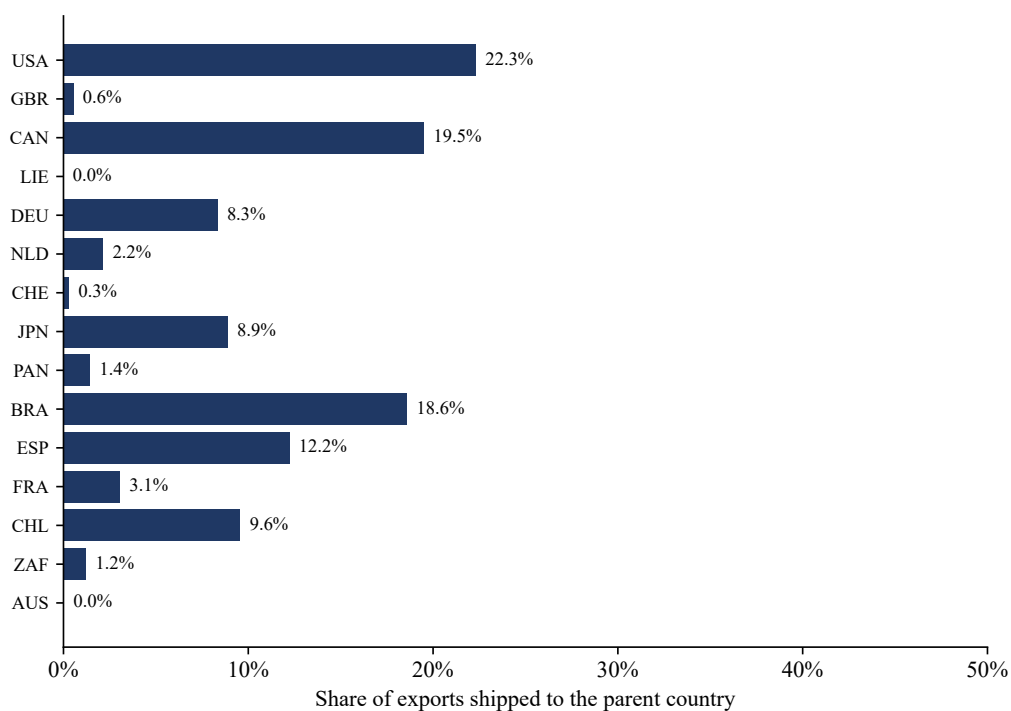


Figure A.6: Share of Foreign-Multinational Exports Shipped to the Parent Country

Notes: Percent of each parent country's exports shipped to that same country.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

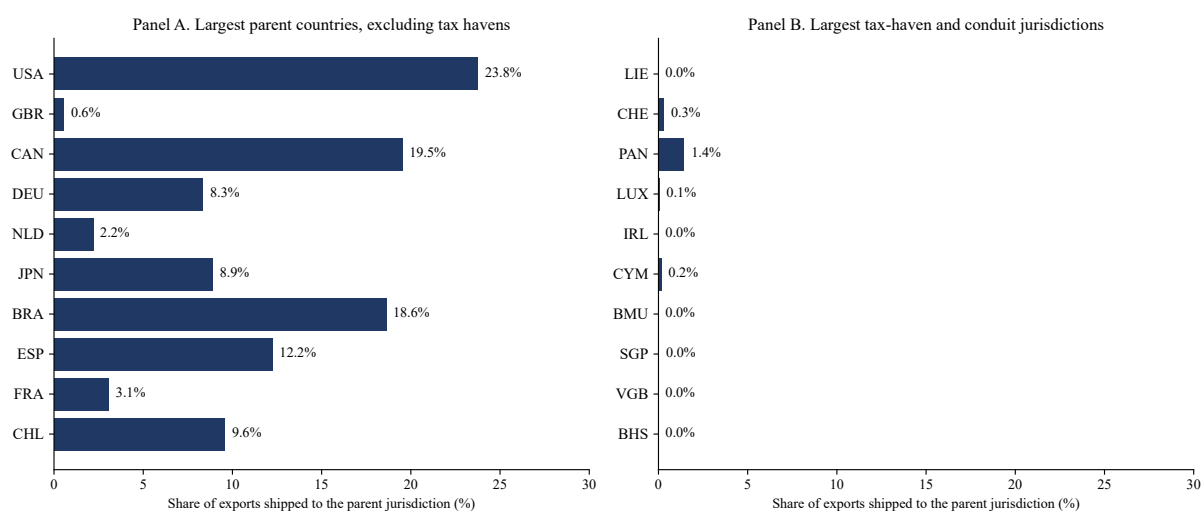


Figure A.7: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction

Notes: Percent of each parent jurisdiction's exports shipped to that same jurisdiction. Panel A: largest parent countries, excluding tax havens; dependencies are assigned to their sovereign state. Panel B: largest tax-haven and conduit jurisdictions.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

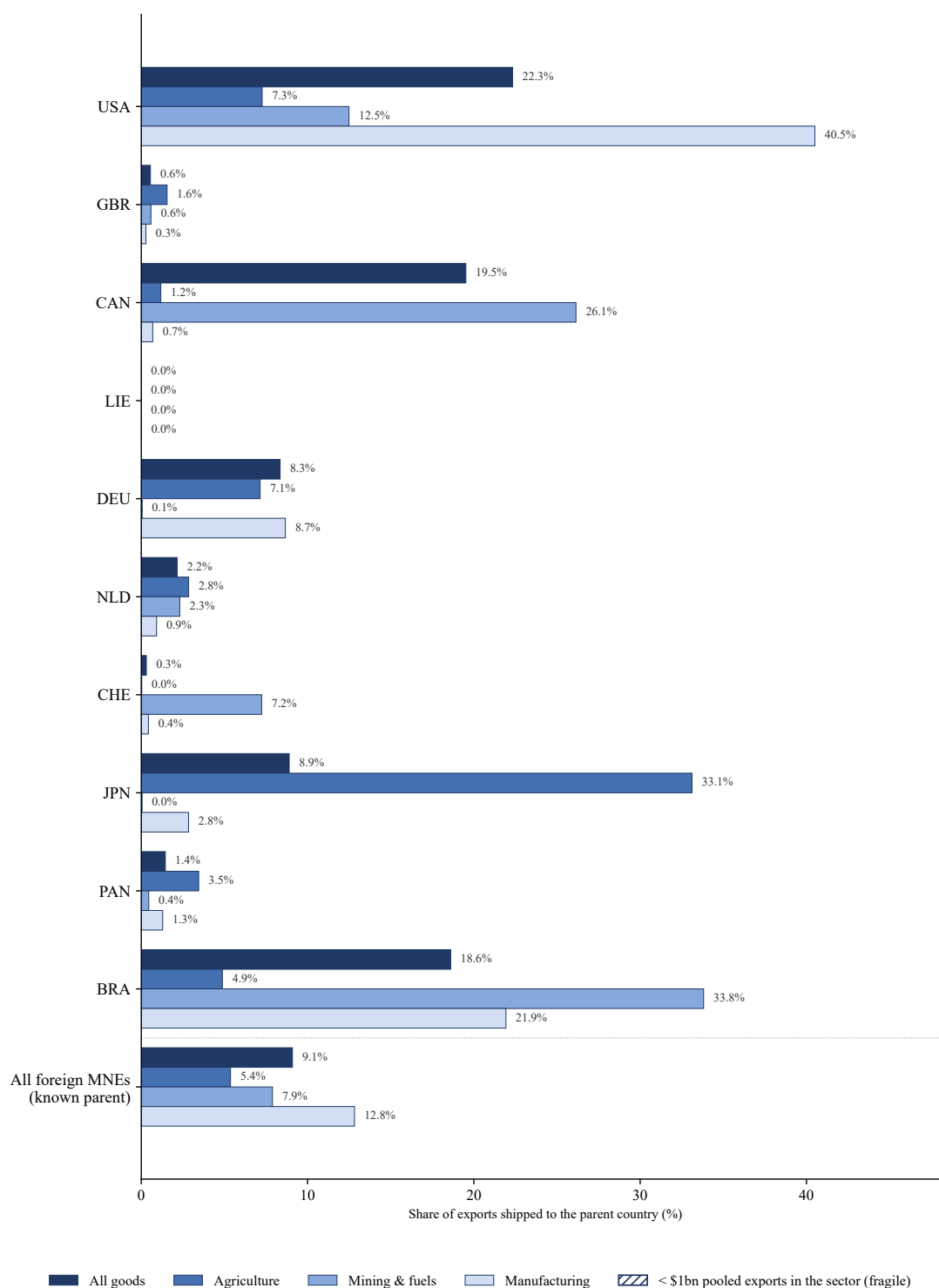


Figure A.8: Share of Foreign-Multinational Exports Shipped to the Parent Country, by Sector

Notes: Percent of each parent country's exports shipped to that same country, by sector. Hatched bars: the parent country's exports in the sector are below US\$1 billion over the period.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

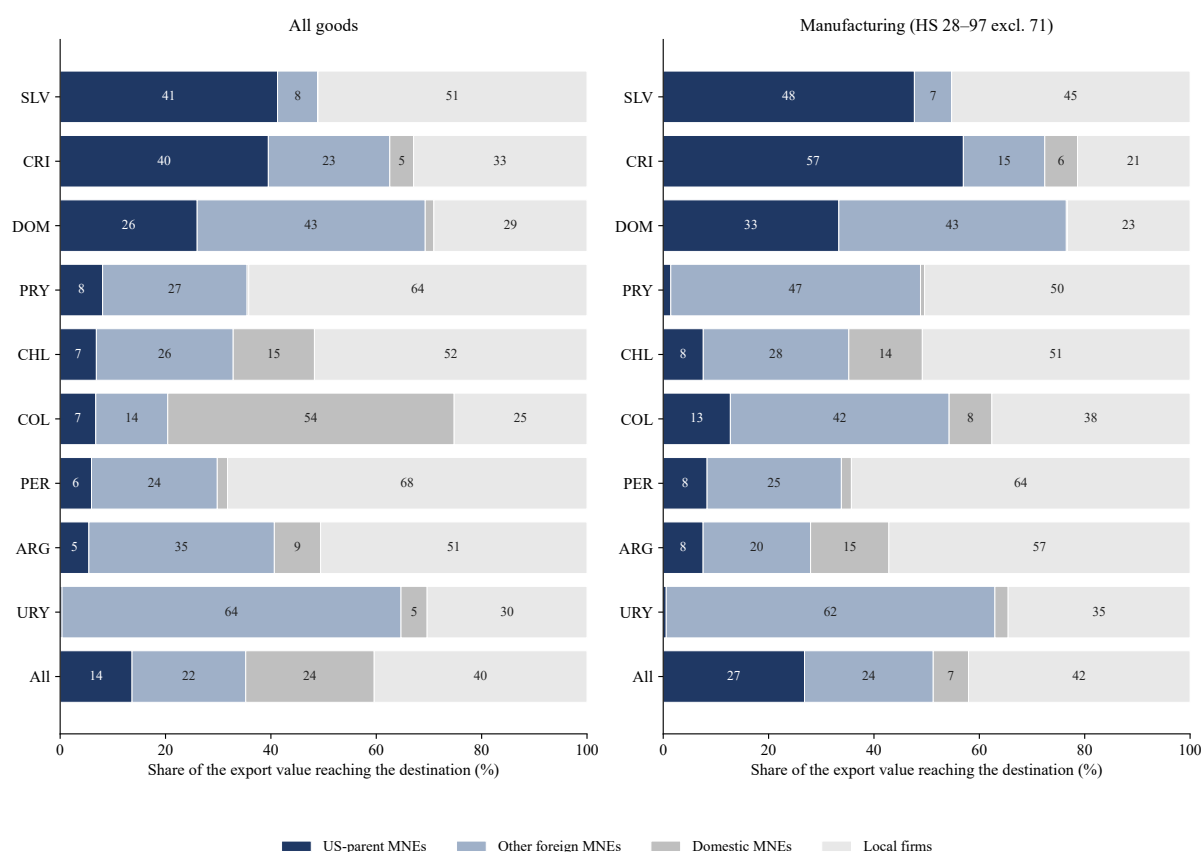


Figure A.9: Exports to the United States, by Type of Exporter and Exporting Country

Notes: Percent of each country's exports to the United States, by type of exporter. Left: all goods. Right: manufacturing.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.6: Exports to the United States, by Type of Exporter and Sector

Sector	Exports to the USA (US\$bn/yr)	US-parent MNEs (%)	Other foreign MNEs (%)	Domestic MNEs (%)	Local firms (%)
All goods	40.0	13.6	21.6	24.4	40.4
Agriculture	10.7	4.8	28.0	9.2	58.0
Mining & fuels	14.6	6.6	14.8	52.0	26.6
Manufacturing	14.5	26.8	24.4	6.7	42.1
Rest	0.1	40.0	38.4	4.5	17.1

Notes: Percent of each sector's exports to the United States, by type of exporter. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.7: Main Products Shipped by US-Parent Multinationals to the United States

HS6	Description	US-parent exports to the USA (US\$bn/yr)	% of all US-parent exports to the USA	% of US-parent exports of the line going to the USA	All LAC exports of the line to the USA (US\$bn/yr)	US-parent share of LAC exports of the line to the USA (%)	Main origins of the US-parent shipments (% of the line)
610910	T-shirts, singlets & other vests, knitted/cro...	0.49	11.1	84	1.01	53	SLV 85%, DOM 10%, PER 5%
901839	Catheters, cannulae & the like	0.52	9.6	73	0.63	89	CRI 92%, DOM 8%, COL 0%
270900	Petroleum oils & oils obt. from bituminous mi...	0.41	9.1	24	7.84	5	COL 100%, ARG 0%
901890	Instruments & appliances used in medical/surg...	0.39	6.8	69	0.94	47	CRI 71%, DOM 29%, COL 0%
270112	Bituminous coal, whether/not pulverised but n...	0.19	4.3	9	0.39	48	COL 100%
854231	Electronic integrated circuits, processors & ...	0.21	3.9	28	0.22	95	CRI 100%, COL 0%, ARG 0%
401110	New pneumatic tyres, of rubber, of a kind use...	0.13	3.4	93	0.24	63	CHL 100%, DOM 0%, COL 0%
261390	Molybdenum ores & concentrates, other than ro...	0.15	2.8	99	0.18	79	PER 100%
611595	Other n.e.s. in 61.15, knitted/crocheted, of ...	0.10	2.4	100	0.11	97	SLV 100%, DOM 0%
610711	Men's/boys' underpants & briefs, knitted/croc...	0.09	2.2	87	0.10	87	SLV 100%, DOM 0%, CRI 0%
621210	Brassieres & parts thereof, whether/not knit...	0.09	1.8	98	0.12	78	SLV 74%, DOM 26%, CRI 0%
080430	Pineapples, fresh/dried	0.08	1.6	66	0.42	20	CRI 100%
240210	Cigars, cheroots & cigarillos containing toba...	0.09	1.3	100	0.57	16	DOM 100%
740311	Cathodes & sections of cathodes, of refined c...	0.06	1.2	24	1.56	3	PER 64%, CHL 36%
640510	Footwear with uppers of leather/composition l...	0.08	1.2	70	0.11	71	DOM 100%
All lines		5.0	100.0	22	40.0	14	

Notes: Fifteen HS6 products with the largest exports by US-parent MNEs to the United States. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Fact 4

Table A.8: Foreign-Multinational Exports and Number of Parents, by Size of the Group's Global Network

Affiliates worldwide	Parents	Share of parents (%)	Share of foreign-MNE export value (%)
1	5,982	48.4	5.4
2–10	3,397	27.5	12.4
11–100	1,893	15.3	26.2
>100	1,099	8.9	56.0
All parents	12,371	100	100

Notes: Number and share of parents, and share of foreign-MNE export value, by the number of affiliates the group has worldwide. Parents with a network record in Orbis or Dun & Bradstreet: 98 percent of parents and 97 percent of foreign-MNE export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Fact 5

Table A.9: Multinational Presence and Exports: Extensive Margin, by Type of Multinational

	All MNEs		Foreign / domestic		Foreign by destination, domestic	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Dependent variable: ln exports, all firms</i>						
Any MNE present	1.4591*** (0.0388)	0.6818*** (0.0165)				
Any foreign MNE			1.4990*** (0.0339)	0.6986*** (0.0158)		
Any foreign MNE, parent-country destination					1.3873*** (0.0951)	0.5909*** (0.0431)
Any foreign MNE, other destination					1.5093*** (0.0319)	0.6976*** (0.0146)
Any domestic MNE			1.4670*** (0.0416)	0.4917*** (0.0165)	1.4334*** (0.0388)	0.4909*** (0.0166)
<i>Panel B. Dependent variable: ln exports, non-MNE firms</i>						
Any MNE present	1.3996*** (0.0367)	0.1311*** (0.0102)				
Any foreign MNE			1.2983*** (0.0346)	0.1356*** (0.0106)		
Any foreign MNE, parent-country destination					1.0183*** (0.0900)	0.1265*** (0.0194)
Any foreign MNE, other destination					1.2782*** (0.0326)	0.1386*** (0.0107)
Any domestic MNE			1.1316*** (0.0428)	0.1370*** (0.0138)	1.0959*** (0.0403)	0.1367*** (0.0138)
Product FE	✓		✓		✓	
Origin × year FE	✓		✓		✓	
Destination × year FE	✓		✓		✓	
Origin × dest. × product FE		✓		✓		✓
Origin × dest. × year FE		✓		✓		✓
Observations (Panel A)	1,829,626	1,633,319	1,829,626	1,633,319	1,829,626	1,633,319
Observations (Panel B)	1,297,111	1,137,550	1,297,111	1,137,550	1,297,111	1,137,550

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: indicators for at least one MNE of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent's country. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.10: Multinational Exports by Exporting Country and Destination Type

Origin	MNE value (US\$bn/yr)	of which domestic (%)	Foreign-MNE value (US\$bn/yr)	% of foreign-MNE value, by destination	
				parent country	other (of which: group affiliate there)
ARG	37.5	13.3	32.5	6.5	93.5 (41.8)
CHL	21.0	22.1	16.4	4.3	95.7 (56.7)
COL	32.8	52.5	15.5	6.5	93.5 (34.2)
CRI	6.6	5.7	6.3	25.5	74.5 (33.8)
DOM	6.0	1.8	5.9	29.0	71.0 (21.6)
PER	19.3	5.0	18.3	3.9	96.1 (28.8)
PRY	4.5	1.8	4.4	2.1	97.9 (65.5)
SLV	2.4	0.6	2.4	40.7	59.3 (28.1)
URY	4.4	15.9	3.7	5.2	94.8 (37.7)
All	134.5	23.4	105.4	8.4	91.6 (40.5)

Notes: MNE and foreign-MNE export value by country, and the split of foreign-MNE value by destination. Parent country: exports to the country of the ultimate parent. Other: all remaining destinations; in parentheses, destinations where the group has another affiliate. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.11: Multinational Presence and Exports by Sector: Intensive Margin

Sample	All goods	Agriculture	Mining & fuels	Manufacturing
<i>Panel A. Dependent variable: ln exports, all firms</i>				
Fixed effects: origin $\times$ year, destination $\times$ year, product				
ln(# foreign MNEs)	1.2129*** (0.0285)	1.2954*** (0.0275)	1.1301*** (0.1799)	1.1644*** (0.0358)
ln(# domestic MNEs)	0.4382*** (0.0259)	0.5233*** (0.0266)	0.3172 (0.2062)	0.3833*** (0.0319)
Observations	113,422	25,437	1,417	85,453
Fixed effects: origin $\times$ destination $\times$ product, origin $\times$ destination $\times$ year				
ln(# foreign MNEs)	0.4506*** (0.0124)	0.4625*** (0.0188)	0.2479 (0.1778)	0.4435*** (0.0148)
ln(# domestic MNEs)	0.2214*** (0.0151)	0.2831*** (0.0234)	0.3216* (0.1639)	0.1904*** (0.0175)
Observations	98,223	22,002	1,027	73,318
<i>Panel B. Dependent variable: ln exports, non-MNE firms</i>				
Fixed effects: origin $\times$ year, destination $\times$ year, product				
ln(# foreign MNEs)	1.1331*** (0.0375)	1.2191*** (0.0363)	0.9420*** (0.2641)	1.0546*** (0.0493)
ln(# domestic MNEs)	0.4707*** (0.0341)	0.5676*** (0.0339)	-0.6594 (0.4830)	0.4194*** (0.0419)
Observations	93,010	22,503	995	68,534
Fixed effects: origin $\times$ destination $\times$ product, origin $\times$ destination $\times$ year				
ln(# foreign MNEs)	0.2413*** (0.0217)	0.3602*** (0.0292)	-0.1512 (0.3320)	0.1865*** (0.0250)
ln(# domestic MNEs)	0.1250*** (0.0218)	0.2461*** (0.0341)	-0.7273 (0.6536)	0.0742*** (0.0238)
Observations	81,563	19,535	726	59,772

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Specification of columns (3)–(4) of Table 9, estimated by sector. Blank: too few observations. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.12: Multinational Presence and Exports by Sector: Extensive Margin

Sample	All goods	Agriculture	Mining & fuels	Manufacturing
<i>Panel A. Dependent variable: ln exports, all firms</i>				
Fixed effects: origin × year, destination × year, product				
At least one foreign MNE exports the product to the destination (0/1)	1.4990*** (0.0339)	1.7972*** (0.0558)	1.4255*** (0.0892)	1.4231*** (0.0344)
At least one domestic MNE exports the product to the destination (0/1)	1.4670*** (0.0416)	1.6133*** (0.0577)	1.7356*** (0.1413)	1.3878*** (0.0466)
Observations	1,829,626	295,788	32,912	1,496,381
Fixed effects: origin × destination × product, origin × destination × year				
At least one foreign MNE exports the product to the destination (0/1)	0.6986*** (0.0158)	0.6929*** (0.0213)	0.7311*** (0.0560)	0.7021*** (0.0178)
At least one domestic MNE exports the product to the destination (0/1)	0.4917*** (0.0165)	0.5683*** (0.0237)	0.5718*** (0.1044)	0.4684*** (0.0186)
Observations	1,633,319	268,332	27,664	1,329,815
<i>Panel B. Dependent variable: ln exports, non-MNE firms</i>				
Fixed effects: origin × year, destination × year, product				
At least one foreign MNE exports the product to the destination (0/1)	1.2983*** (0.0346)	1.7649*** (0.0546)	1.5696*** (0.1099)	1.1402*** (0.0344)
At least one domestic MNE exports the product to the destination (0/1)	1.1316*** (0.0428)	1.3082*** (0.0539)	1.1333*** (0.1497)	1.0491*** (0.0490)
Observations	1,297,111	232,617	22,836	1,038,206
Fixed effects: origin × destination × product, origin × destination × year				
At least one foreign MNE exports the product to the destination (0/1)	0.1356*** (0.0106)	0.2854*** (0.0179)	0.1991*** (0.0547)	0.0997*** (0.0114)
At least one domestic MNE exports the product to the destination (0/1)	0.1370*** (0.0138)	0.2610*** (0.0201)	0.2125*** (0.0899)	0.0984*** (0.0167)
Observations	1,137,550	208,479	18,666	904,114

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Specification of columns (3)–(4) of Table A.9, estimated by sector. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Fact 6

Table A.13: Distance and Firm Exports, by Sector

Dep. var.: ln firm exports	All goods	Agriculture	Mining & fuels	Manufacturing
Origin, destination, year and product FE				
ln distance	-0.1645*** (0.0454)	-0.2135*** (0.0748)	-0.0926 (0.0819)	-0.1510*** (0.0454)
ln distance × foreign MNE	0.0487*** (0.0041)	0.0366*** (0.0034)	0.0566*** (0.0086)	0.0515*** (0.0048)
ln distance × domestic MNE	0.0318*** (0.0042)	0.0187*** (0.0046)	0.0706*** (0.0180)	0.0388*** (0.0052)
Observations	5,081,547	1,045,009	79,938	3,917,960
Origin × year, destination × year and product FE				
ln distance	-0.1853*** (0.0460)	-0.2285*** (0.0711)	-0.1001 (0.0828)	-0.1742*** (0.0459)
ln distance × foreign MNE	0.0483*** (0.0040)	0.0358*** (0.0034)	0.0557*** (0.0087)	0.0512*** (0.0048)
ln distance × domestic MNE	0.0318*** (0.0042)	0.0182*** (0.0046)	0.0673*** (0.0183)	0.0390*** (0.0052)
Observations	5,081,360	1,044,787	79,612	3,917,765

Notes: Observations are firm–destination–HS6–year cells. Dependent variable: ln exports. Specification of columns (1)–(2) of Table 11, estimated by sector. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Products

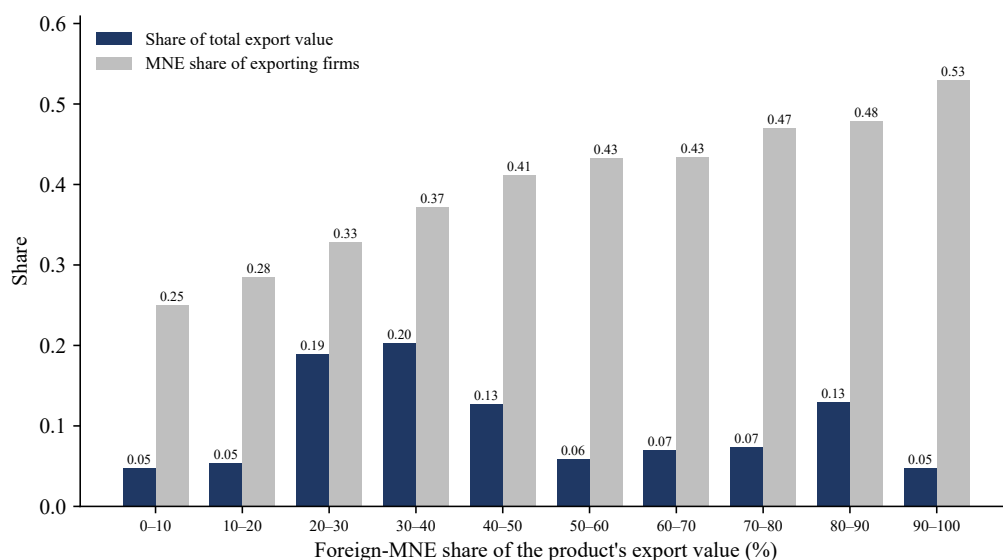


Figure A.10: Distribution of Export Value by Product-Level Foreign-Multinational Share

Notes: Share of total export value and MNE share of exporting firms, by the foreign-MNE share of each HS6 product's export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.14: Distribution of Export Value by Product-Level Foreign-Multinational Share

Foreign share bin (%)	Share of value	MNE share of exporting firms	HS6 products
0-10	0.048	0.250	1,426
10-20	0.054	0.285	731
20-30	0.189	0.329	652
30-40	0.202	0.372	591
40-50	0.127	0.411	519
50-60	0.059	0.433	442
60-70	0.070	0.435	386
70-80	0.074	0.470	311
80-90	0.129	0.479	311
90-100	0.048	0.530	560

Notes: Share of total export value and MNE share of exporting firms, by the foreign-MNE share of each HS6 product's export value. MNE share of exporting firms: MNE firm-destination-product-year observations over all observations in the bin.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.15: Agriculture: Exports and Multinational Shares, by SITC Division

SITC Rev. 3 division	Total exports (US\$bn/yr)	MNE exports (US\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
08 Animal feed	13.8	6.6	35.7	11.5	CHE 0.28, LIE 0.19, USA 0.18
05 Vegetables & fruit	10.6	4.0	32.6	5.8	ESP 0.14, USA 0.13, PAN 0.11
04 Cereals	10.3	5.9	48.6	8.2	USA 0.33, CHE 0.14, NLD 0.14
22 Oil seeds	6.7	4.4	61.6	4.1	USA 0.54, NLD 0.14, LIE 0.08
07 Coffee, tea, cocoa, spices	5.1	2.3	30.4	15.6	DEU 0.41, ITA 0.14, SGP 0.12
42 Fixed vegetable oils	6.0	2.5	30.8	10.2	CHE 0.27, NLD 0.24, USA 0.23
01 Meat	5.2	2.4	34.8	13.1	SAU 0.46, BRA 0.26, JPN 0.08
03 Fish & seafood	3.4	1.8	41.9	9.5	ESP 0.32, LUX 0.11, ITA 0.10
29 Crude animal/veg. materials	2.2	0.8	27.3	9.0	DEU 0.10, ARG 0.09, BGR 0.08
11 Beverages	2.2	0.8	28.6	9.2	LUX 0.17, PAN 0.15, GBR 0.12
02 Dairy & eggs	1.9	1.0	28.0	20.3	NLD 0.20, CHE 0.19, ROU 0.19
09 Misc. edible products	1.7	1.2	70.9	2.3	USA 0.46, CHE 0.17, FRA 0.09
06 Sugar & honey	1.4	0.5	19.3	14.7	USA 0.50, KOR 0.18, LIE 0.10
12 Tobacco	1.2	0.7	54.8	0.2	USA 0.36, HKG 0.25, GBR 0.12
41 Animal oils & fats	0.5	0.2	23.0	11.0	BMU 0.31, PAN 0.23, GBR 0.16
Other	0.6	0.1	21.6	1.9	CHE 0.21, USA 0.20, ROU 0.10
All	72.9	35.0	38.5	9.4	

Notes: Agriculture: HS 01–24. Foreign and domestic: percent of the division's export value. HS6 products without an SITC correspondence (US\$6.0 billion a year) are excluded. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.16: Manufacturing: Exports and Multinational Shares, by HS Section

HS section	Total exports (US\$bn/yr)	MNE exports (US\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
VI Chemicals	11.6	7.4	49.3	14.0	USA 0.25, DEU 0.13, GBR 0.12
VII Plastics and rubber	5.8	3.4	50.5	7.7	USA 0.34, MEX 0.17, JPN 0.06
VIII Hides, skins, leather	1.6	0.8	39.7	11.6	BGR 0.19, URY 0.17, SGP 0.16
IX Wood and articles	1.9	1.1	27.4	33.9	BHS 0.24, JPN 0.12, VGB 0.11
X Pulp, paper	3.4	2.7	54.2	28.2	PAN 0.54, USA 0.19, CHL 0.08
XI Textiles and apparel	6.9	3.6	49.1	2.9	USA 0.62, CAN 0.04, FRA 0.03
XII Footwear, headgear	0.5	0.3	48.2	1.3	USA 0.66, BRA 0.13, JPN 0.08
XIII Stone, ceramics, glass	0.9	0.5	49.5	4.5	USA 0.32, PAN 0.20, CYM 0.11
XV Base metals	19.6	8.5	41.1	3.1	GBR 0.44, CAN 0.12, ZAF 0.08
XVI Machinery and electrical equipment	6.2	4.2	63.2	3.9	USA 0.41, IRL 0.10, JPN 0.08
XVII Transport equipment	9.7	7.9	77.6	4.2	DEU 0.30, JPN 0.28, NLD 0.12
XVIII Optical, precision instruments	3.4	2.8	72.3	8.5	USA 0.66, DEU 0.12, GBR 0.08
XX Miscellaneous manufactures	0.9	0.6	52.9	6.9	USA 0.48, SWE 0.10, DEU 0.10
Other	0.1	0.0	38.0	0.5	BGR 0.92, USA 0.03, CHE 0.01
All	72.7	43.9	51.6	8.0	

Notes: Manufacturing: HS 28–97 excluding 71. Foreign and domestic: percent of the section's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.17: Manufacturing: Exports and Multinational Shares, by End Use

BEC end use	Total exports (US\$bn/yr)	MNE exports (US\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
Intermediate	47.2	26.6	46.9	9.1	GBR 0.20, USA 0.20, PAN 0.07
Consumption	15.2	10.0	60.6	4.9	USA 0.43, NLD 0.09, DEU 0.08
Capital	9.3	6.7	64.8	5.6	DEU 0.31, JPN 0.28, USA 0.18
All	71.7	43.3	51.7	7.8	

Notes: End use from BEC Rev. 4. Foreign and domestic: percent of the category's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table A.18: Mining and Fuels: Exports and Multinational Shares, by End Use

BEC end use	Total exports (US\$bn/yr)	MNE exports (US\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
Intermediate	65.4	48.6	51.6	23.7	GBR 0.37, USA 0.21, CAN 0.16
Consumption	0.5	0.3	61.7	0.4	USA 0.22, FRA 0.22, BEL 0.21
Capital	0.0	0.0	21.4	0.0	FRA 0.82, USA 0.08, GBR 0.03
All	65.9	48.9	51.6	23.6	

Notes: Mining and fuels: HS 25–27 and 71. End use from BEC Rev. 4. Foreign and domestic: percent of the category's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.